# INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

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(Affiliated to UEM Kolkata and Approved by AICTE)

## Degree: BCA, Stream: BCA

## Term – I , Third Semester Examination, September - 2024

## Subject Code: BCAAE301

## Subject Name: Python Programming

Full Marks: 30 Duration: 1 Hour

**Part - A**

**Attempt 5 questions**

**Each question carries 2 marks** (2 × 5)

1. What is List in Python? How can we define a List in Python? Give an example of a List containing 5 string type data. 2

**or**

With the help of an example show how to create a String in Python? Write a Python program to reverse a string using negative indexing. 2

2. Write the difference between List and Tuple. With the help of an example show how to create a List and Tuple in Python? 2

**or**

Write a Python program to display even numbers from a list of elements. 2

3. .Write the output of the following 2

```
str = "Hello, Student!"
print(b[3:6])
```

**or**

Write a Python program to Append new string in the middle of a given strings: 2

Input: s1 = "Ault" s2 = "Kelly“

Output: AuKellylt

4. Write the difference between break and Continue keywords in a loop 2

**or**

Explain the syntax range function a loop. 2

5. Write the syntax of for loop and analyse it. 2

**or**

Find the output of the following: 2

```
If not True:
    print(“10”)
```

```
elif not(10+10 == 30):
    print("20")
else:
    print("30")
```

## Part - B

### Attempt 2 questions

### Each question carries 5 marks (5 × 2)

6. Write a program in python which will input a year and find whether it is a leap year or not. **5**

**or**

Write a program to get the principal amount, interest and year and find the simple and compound interest. **5**

7. Write a Python program to calculate the GCD of two numbers. Take both numbers as user input. Print the common factors of the two numbers as well. **5**

**or**

Write a Python program to print the Fibonacci Series using loop only for $n$ series. Use of functions are prohibited. Take the value of $n$ as user input. **5**

## Part - C

### Attempt 1 question

### Each question carries 10 marks (10 × 1)

8. Write a program to print the sum of the following $n^{th}$ series. **10**

$$S = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots n$$

**or**

Write a program to print the following pattern for $n$ rows **10**

```
  *
 * *
*   *
 * *
  *
```